Oliver Neal

Technical Director, Shiverbug Studios · Unity (C#)

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PROFILE

Technical Director at Shiverbug Studios, leading the technical side of *Out of Water*, a two-player co-op 3D platformer in Unity 6. My own work is the systems layer: state-machine-driven characters and enemy AI, local co-op and multiplayer plumbing, tools for the team, and performance work that is measured before it is claimed.

First Class Honours in BSc (Hons) Games Development from Teesside University, and winner of Best Games Programming and Development Artefact at ExpoTees 2026. Nine projects before the studio, most playable in the browser, with source on GitHub and breakdowns at oliverneal.dev. Now studying an MA in Games Design alongside the studio role.

EXPERIENCE

Technical Director · Shiverbug Studios — Oct 2026 to present

- Lead the technical side of the studio and sit on its board, alongside the MA. I write most of the studio's code, on a team of over a dozen.
- Added online co-op alongside splitscreen, tested at a simulated 150 ms of lag before it reaches a playtest.
- Split the code into core, game, online and UI assemblies, which took gameplay's references to the online session from 1,022 to none, and put the moveset and combat under automated tests.
- Took 15 to 18% of the draw calls out of splitscreen on the lowest quality tier.
- Own the build (git workflow, pull request review, a tagged playtest build every fortnight), the technical plan for the vertical slice, and the studio's repositories and access.
- Get artists and designers working confidently in Unity and git, so an art-only change never has to wait for a programmer.

Gameplay Programmer (Internship) · Shiverbug Studios — Jul to Sep 2026

- Hired for the character and enemy gameplay code on *Out of Water*, an original-IP two-player co-op 3D platformer; by the end of the ten weeks I was also the studio's buildmaster.
- Rebuilt both playable characters and the enemies from single-file controllers onto component-driven finite and hierarchical state machines, and built the co-op moveset on top.
- Ran the performance pass: quality tiers, LOD and culling, measured branch to branch with a profiling tool I made for the team. Trialled Unity's GPU Resident Drawer and turned it back off when it cost more than it saved.
- Built the splitscreen plumbing (per-player device routing, cameras, character swapping) and a UI Toolkit front end: HUD, character select, pause, save slots and a controls tutorial.

Also: water shaders in Shader Graph and URP HLSL, a decoupled FMOD audio layer, level design tooling, and the studio's website, shiverbugstudios.com.

TECHNICAL SKILLS

Languages & engines: C#, Unity 6 (2D/3D), Shader Graph, HLSL, Unreal Engine 5

Gameplay & AI: gameplay programming, systems design, finite & hierarchical state machines, AI navigation (NavMesh), local co-op & splitscreen, Unity Input System

Performance: profiling and frame analysis, time-sliced & multi-rate simulation, LOD and culling, quality tiers, runtime NavMesh baking

Multiplayer & services: Netcode for GameObjects, Unity Authentication, Relay, Cloud Save

Audio, UI & presentation: FMOD, UI Toolkit, Unity UI, Cinemachine, particle VFX

Tools & workflow: Git and Git LFS, branch workflow and pull request review on a shared team repo, automated testing (Unity Test Framework), Visual Studio, JetBrains Rider

EDUCATION

MA Games Design (with Advanced Practice) · Teesside University · in progress — Sep 2026 to 2028

BSc (Hons) Games Development · Teesside University · First Class Honours, 80% overall — Jul 2026

FdSc Games Technologies · Newcastle College University Centre · 2:1 — Jul 2025

BTEC Level 3 National Foundation Diploma · Creative Digital Media Production · Newcastle College · Distinction — 2023

SELECTED PROJECTS

Cyber Station · Unity 3D · C# · AI navigation · simulation · solo — Apr 2026

- Neon station management game, 86 scripts and around 17,000 lines: hundreds of passengers run off one fixed 10 Hz tick, picking facilities by lowest estimated total delay (queue wait plus real NavMesh walk time, both in seconds), with six live ratings feeding back into the spawn rate.
- Final-year artefact: scored 90/100 and won Best Games Programming and Development Artefact at ExpoTees 2026.

Breakdown: oliverneal.dev/cyber-station.html · Play: olivernealdev.itch.io/cyber-station · Source: github.com/OliverNealDev/CyberStation

Minimalists · Unity 3D · C# · enemy AI · RTS · solo — Aug 2025

- Node-conquest RTS with six enemy AI personalities and no extra information for any of them: one scores every source and target pair by distance and garrison, and all six commit through the same send call the player's slider drives.

Breakdown: oliverneal.dev/minimalists.html · Play: olivernealdev.itch.io/minimalists · Source: github.com/OliverNealDev/Minimalists

Wanted · Unity 6 · C# · stealth AI · one-week game jam · solo — May 2025

- Top-down stealth survival game built in one week for the Tyne to Game jam: one detection meter that fills four times slower under cover, three-state officer AI that searches your last known position, and a police force that grows car by car, on a runtime-baked 2D NavMesh.

Breakdown: oliverneal.dev/wanted.html · Play: olivernealdev.itch.io/wanted · Source: github.com/OliverNealDev/Wanted

Unicellular · Unity · C# · FSM · optimisation · solo — Feb 2025

- Idle simulation stress-tested at 1,000 and 10,000 concurrent entities, kept smooth by a decoupled multi-rate tick and time-sliced proximity AI.

Breakdown: oliverneal.dev/unicellular.html · Play: olivernealdev.itch.io/unicellular · Source: github.com/OliverNealDev/Unicellular

Also

Tile Turfer (Unity 2D, Jan 2026), 3D Netcode Tank Game (Unity 3D, Feb 2025) and Pinnable (Unity 2D, Mar 2024), all solo. In teams: Space Bar Simulator (Unreal Engine 5, team of 10, Jan 2026), where I owned the player controller, the interaction system and the camera-blended stations, and Candy Corporate (Unity 2D, Apr 2024), where I did level design and pixel art integration. Breakdowns for all but the last are at oliverneal.dev.